

Assessment Tool 3 - Elaborate Answers

Q1. Smart home: Explain simple reflex, model-based, goal-based and utility-based agents with examples. Simple reflex responds to current inputs such as turning lights on when motion is detected. Model-based agents maintain an internal model of occupancy and environment. Goal-based agents plan actions to achieve comfort, security and energy-saving goals. Utility-based agents evaluate multiple choices and select the action with maximum utility by balancing comfort, electricity usage and security. A hybrid approach combining model-based and utility-based agents is the most suitable because it reacts quickly, learns preferences, predicts situations and optimizes overall performance. Advantages include adaptive behaviour, energy efficiency, user comfort and improved security.

Q2. Uniform Cost Search expands the node with the lowest accumulated cost and guarantees the optimal least-cost path when costs are non-negative, but it consumes high memory. Iterative Deepening Search repeatedly performs depth-limited search with increasing depth, combining the low memory of DFS with the completeness of BFS. UCS has high time and space requirements whereas IDS has time $O(b^d)$ and space $O(bd)$. A model-based goal agent is suitable because it updates knowledge of the maze while searching for the exit. When path costs differ, UCS is preferred; when memory is limited and costs are equal, IDS is more practical. Therefore, a model-based goal agent with UCS provides the best solution for variable-cost environments.